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Member 4

Dataset Evaluation and AI Workflow

Documentation

Objective

The objective of today's task was to study available virtual try-on datasets and identify the most suitable datasets for the Raritone platform. Additionally, a complete AI workflow was designed to understand how a user's image is processed and transformed into a realistic virtual try-on output.

Approach Followed

Instead of only listing datasets, I evaluated them from the perspective of Raritone's requirements.

The evaluation focused on:

- Image quality
- Clothing categories available
- Pose diversity
- Suitability for virtual try-on systems
- Ease of integration into AI models

The goal was to identify datasets that can support future development of a realistic and scalable fashion try-on platform.

Dataset Evaluation

1. VITON

Purpose:

2D Virtual Try-On

Advantages:

- Widely used benchmark dataset
- Suitable for clothing transfer tasks
- Easy to implement for research purposes

Limitations:

- Limited image resolution
- Less realistic compared to newer datasets

Suitability:

Good for initial experimentation.

2. VITON-HD

Purpose:

High-resolution Virtual Try-On

Advantages:

- High-quality images
- Better garment details
- More realistic results
- Popular in recent virtual try-on research

Limitations:

- Requires higher computational resources

Suitability:

Highly suitable for Raritone.

3. Dress Code

Purpose:

Multi-category Clothing Dataset

Advantages:

- Contains dresses, tops, pants and multiple clothing categories
- More diverse clothing combinations
- Suitable for real-world applications

Limitations:

- Larger dataset requires additional preprocessing

Suitability:

Excellent choice for Raritone.

4. DeepFashion

Purpose:

Fashion Recognition and Clothing Analysis

Advantages:

- Large-scale fashion dataset
- Rich attribute information
- Useful for recommendation systems

Limitations:

- Not designed exclusively for virtual try-on

Suitability:

Useful for both recommendation and try-on modules.

5. MPV Dataset

Purpose:

Multi-Pose Virtual Try-On

Advantages:

- Multiple body poses
- Helpful for pose-aware virtual try-on systems

Limitations:

- Smaller compared to newer datasets

Suitability:

Good for pose variation studies.

6. FashionGen

Purpose:

Fashion Recommendation Dataset

Advantages:

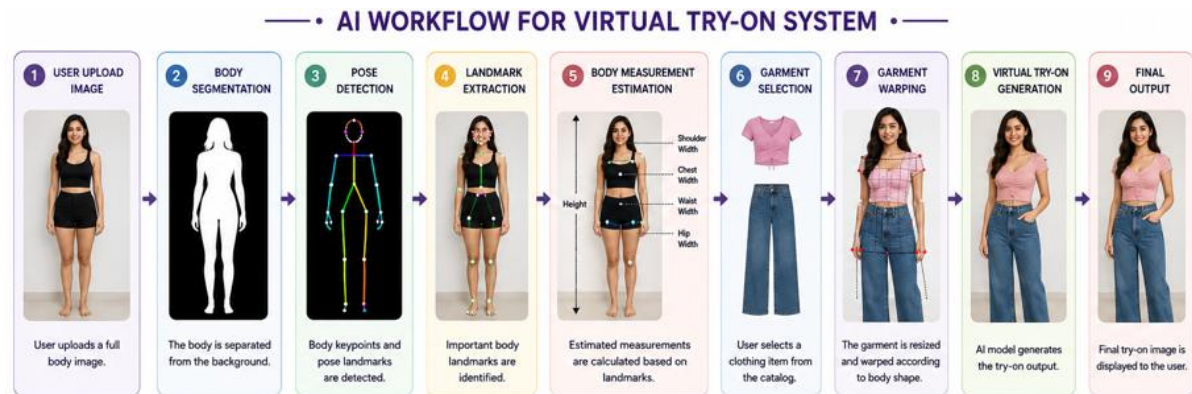
- Rich fashion descriptions
- Useful for AI-based recommendations

Limitations:

- Limited virtual try-on support

Suitability:

Best suited for recommendation engines.



Recommended Datasets for Raritone

After evaluation, the following datasets are recommended:

Primary Dataset

VITON-HD

Reason:

Provides high-quality images and realistic garment transfer results.

Secondary Dataset

Dress Code

Reason:

Offers diverse clothing categories suitable for real-world fashion applications.

Supporting Dataset

DeepFashion

Reason:

Useful for clothing recognition, recommendation systems and fashion attribute extraction.

Proposed AI Workflow for Raritone

Step 1 – User Image Upload

User uploads a full-body image through the application.

Step 2 – Body Segmentation

The user's body is separated from the background.

Step 3 – Pose Detection

Body keypoints and pose landmarks are detected.

Step 4 – Landmark Extraction

Important body landmarks such as shoulders, waist, hips and knees are identified.

Step 5 – Body Measurement Estimation

Approximate body measurements are calculated.

Step 6 – Garment Selection

The user selects a clothing item from the catalog.

Step 7 – Garment Warping

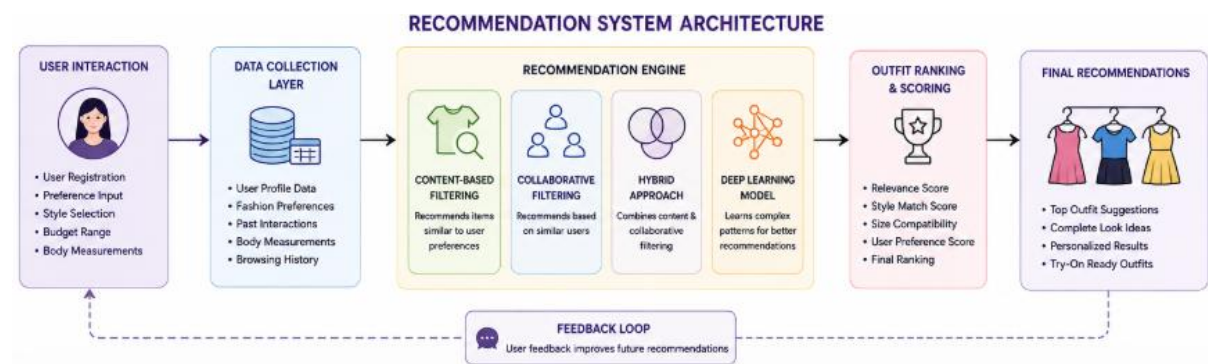
The selected garment is reshaped according to the user's body structure.

Step 8 – Virtual Try-On Generation

The AI model combines the garment and user image.

Step 9 – Final Output

The final virtual try-on image is displayed to the user.



Findings

- VITON-HD provides the best image quality for virtual try-on tasks.
- Dress Code improves clothing category diversity.
- DeepFashion can support both recommendation and try-on modules.
- Body segmentation and pose detection are critical for realistic garment fitting.
- Combining multiple datasets can improve overall system performance.

Conclusion

The study identified VITON-HD, Dress Code and DeepFashion as the most suitable datasets for the Raritone platform. A complete AI workflow was also designed to demonstrate how a user's image can be processed from upload to final virtual try-on generation. This workflow can serve as the foundation for future development of an AI-powered fashion recommendation and virtual try-on system.